

Repressive Democracies: the Relationship Between State's Repression Severity, Protests, and Policy Change.

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September 5, 2026

1 Introduction.

Scholarly work that inquiries into state repression's consequences on mobilization dynamics have revealed puzzling results. On the one hand, some studies show that threats of state repression may deter further mobilization by increasing its cost to protesters. In opposition, others reveal that repression backfires, meaning that protesters who are victims of violence become radicalized. Moreover, when one thinks of repression severity as a policy variable that the states choose, it becomes clear that repression and dissent are strategically interlinked.

The study of this phenomenon, framed as the “protest-repression” nexus, has been developed primarily in the context of autocracies. This is especially true when theoretical or micro-founded approaches have been put forth to explain empirical patterns in cross-country perspectives. Yet, there are several reasons to think that this phenomenon deserves a closer and deeper study in democratic regimes. First, democracies are naturally well-suited environments for individuals that consider protesting if they hold grievances against political elites or other societal groups.

Second, in democracies, for incumbent political elites, mass protests are costly for two reasons. On the one hand, a massive protest may force politicians to change policies they might not want to alter. For example, gridlocked constitutional dispositions that benefit elites, such as the tax code,

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the electoral system, or others. Additionally, massive protests imply that the state needs to spend resources to “oversight” them. Most of the time, this implies having a specialized anti-protest or anti-riot police corps, which possibly implies expensive budgets to sustain them.

In democracies, the severity of repression is a complex policy that is implicitly written in the institutional and legal framework. For democratic regimes, leaders do not command repressive bureaus to repress citizens in particular ways. Nevertheless, defining a given legal structure that allows repressive operatives to exert higher or lower degrees of physical or physiological damage to dissenter citizens implicitly sets the severity of the repression policy. Naturally, we would think that this policy is hard to quickly modify if a protest event has already started. Repressive bureaus likely know the rules of engagement they can follow, imprinted on the formal institutional framework, which allows them to use different levels of violence on citizens. Since savvy politicians know that protests are always possible in the foreseeable future, they will think very hard when setting the repression severity policy, weighing costs and benefits.

In this paper, I develop a game-theoretic model that integrates the state’s decision regarding the severity of the repression policy and the citizens’ decision to protest motivated by a prospect of policy change. The model has a number of essential features. First, citizens face a collective action problem when they decide to protest: the protest changes the policy only if participation reaches a critical threshold. Second, that critical threshold is not common knowledge. Citizens hold noisy private assessments of how large the protest must be to force the policy change, and they must decide whether to participate under this strategic uncertainty. This is the global games structure: as in the canonical regime change game, the incomplete information about the fundamental delivers a unique equilibrium in threshold strategies, where the complete-information version of the same game exhibits multiple equilibria. Third, given that the state would prefer to keep the status quo, citizens’ and state incentives are directly opposed. Thus, the state decides on a repression severity policy that imposes costs on individuals who choose to protest. Importantly, although a higher severity of repression may deter protests, setting a repression severity policy is costly for the state.

The rest of the paper is organized as follows. First, I offer a brief literature review that informs the assumptions and context of the paper, including the global games literature on protest and regime change that supplies the machinery for the citizens’ stage. Next, I develop the citizens’

participation decision as a global game of policy change, presenting the complete-information benchmark first to make the role of the incomplete information explicit. Then, I integrate the state's decision on the repression severity policy into a two-stage sequential game. Furthermore, I discuss the implications of the model, underlining insights for empirical research. Finally, I present the conclusions of the paper.

2 Literature review.

2.1 State repression, political attitudes, and social movements.

The study of state repression has a long and vast literature. Broadly, the act of subduing or coercing individuals or groups that hold claims to or challenge political actors or the status quo, using institutional or physical force, most commonly using violent means, is known as political or “state” repression (deMeritt 2016). Many empirical regularities have been documented and different theoretical approaches have been utilized to understand why states repress citizens, and how this influences their political behavior, especially their participation in protests. It should not be surprising that this body of research is overwhelmingly focused on authoritarian regimes or flawed democracies. The intrinsic characteristics of such regimes, such as the lack of accountability of political leaders, and the absence of freedom of speech or competitive elections, enable rulers to deter opposition using violent means.

Thus, comparative research persistently revealed that democratic countries exhibit lower levels of state repression (Davenport 2007a; Davenport and Armstrong 2004). Yet, state repression, especially in the form of violent coercion perpetrated by state agents, or “repressive operatives,” is not uncommon in developed democracies. In fact, state violence, especially police brutality, has become increasingly ubiquitous across some democracies (Belur 2010; Whitehead, Fair, and Payne 2010; Worden 2015). These expressions of violence tend to be very salient and capture the attention of the public. For example, there is evidence that after the killing of George Floyd at the hands of the police, support for the police substantially fell (Reny and Newman 2021). Moreover, victims of police repression may experience a range of emotional states such as anxiety, depression, anger, and fear (Bor et al. 2018; Çelebi, Adam-Troian, and Mahfud 2020; Cooper et al. 2004; J. E. DeVlyder et al. 2017; J. DeVlyder, Fedina, and Link 2020).

Yet not all citizens perceive the use of force by repressive bureaus in the same fashion. For example, in the U.S., studies have persistently revealed that conservatives have a propensity to support the police at higher rates than liberals (Ekins, 2016; Fingerhut, 2017; Hindelang, 1974; Zamble & Annesley, 1987). To the extent that voters have directional motivations and are affective partisans, they may perceive, process, and evaluate state repression in different ways. For example, attitudes about the police are racially divided (Ekins, 2016; Fingerhut, 2017).

One essential question in the literature is what the consequences of state repression in the context of social movements are. Studies have generally referred to this problem as the “punishment puzzle” (Davenport 2007b). On the one hand, studies have shown that repression discourages mass and peaceful protests but stirs radicalization (della Porta 1995). In contrast, there is also evidence that suggests that repression incites conflict escalation and radicalization, thus provoking higher levels of dissent and mobilization (Adam-Troian, Çelebi, and Mahfud 2020; Christopher Michael Sullivan and Davenport 2017).

2.2 Empirical regularities and theoretical foundations.

The literature has studied several determinants of state repression, revealing that regime type, economic development, population size, intrastate civil conflicts, and degrees of grievances and dissent systematically explain state repression (Davenport 2007b; Nordås and Davenport 2013; Christopher M Sullivan 2016; Christopher Michael Sullivan and Davenport 2017). For example, Nordås and Davenport (2013) argue that the demographic structure of the population is a key determinant of the level of state repression. To the extent that countries exhibit larger cohorts of “youth bulges”, people between the ages 15–24, relative to the overall adult population, then states respond by increasing repression. The rationale for this response is that if states anticipate a higher level of dissent, then they preemptively increase repression to induce deterrence.

Moreover, Chenoweth, Perkoski, and Kang (2017) summarize several consistent findings in the literature. To begin with, a systematic finding is that dissent antecedes state repression in most settings. This finding has been labeled as the “law of coercive responsiveness” (Davenport 2007b). Also, democratic countries tend to exhibit lower levels of repression. Note that the authors also discuss the “punishment puzzle” mentioned above. Overall, they stress that repression and dissent, in the form of protests or other tactics, are two complex interlinked processes.

On the side of formal models, several approaches have been attempted to shed light on the theoretical underpinnings of the relationship between state repression, protests, and regime or policy change. Although there is not a definite consensus for what is the best modeling approach. Ritter (2014) notes that models have chosen different main objects of investigation that, if taken separately, miss some of the key strategic features of the relation between repression and dissent. For example, some studies model the determinants of regime change or dissent in general as a process of an ongoing or expected conflict. One way to approach this is that citizens to protest to alter the status quo, which depends on the state's concessions. For example, the level of violence of a revolutionary vanguard may signal the level of dissent in society, such that citizens' cooperation is provoked (Bueno de Mesquita 2010). Other formal models have stressed the role of uncertainty and asymmetries of information in government types (Pierskalla 2010). For example, a "strong" state may be willing to costly repress citizens to signal its resolve. Over repeated play, citizens protests and learn about the type of government they face.

Thus, as Ritter (2014) notes, some models study the conditions that ignite dissent, and others study how the state chooses the severity of repression. But, if these studies take these two processes as separate entities, then an important part of the puzzle is missed. Hence, she proposes a model that jointly considers the conditions that spark dissent, or protest, and how the state chooses the severity of repression. The strategic setting is modeled by a bargaining model of policy provision, and its effect on political incumbent survival. The results show that an increase in the likelihood of survival of the political leaders decreases the level of dissent and repression. Yet, an increase in the likelihood of survival of the political leaders increases the severity of repression and the severity of dissent.

2.3 Global games of protest and regime change.

The machinery this paper borrows comes from the global games literature on coordination under incomplete information (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). In the canonical global game of regime change, a continuum of citizens simultaneously decide whether to attack a regime that collapses if and only if the mass of attackers exceeds the regime's strength, which is an uncertain fundamental about which citizens receive noisy private signals. Under complete information about the fundamental, the game exhibits multiple equilibria in an intermediate

range of regime strength; the incomplete information selects a unique equilibrium in threshold strategies.

Two recent applications inform the present model directly. Bueno de Mesquita and Shadmehr (2023) develop “a standard global game of regime change” in which a continuum of citizens receive noisy private signals $x_i = \theta + \sigma \varepsilon_i$ about the regime’s strength θ , and the rebellion succeeds if and only if the fraction of rebels exceeds θ . Their complete-information benchmark (their Proposition 1) shows that for intermediate strength “both equilibria coexist,” and they note that this multiplicity “makes empirical prediction difficult” and renders outcomes “insensitive to parameters of the model like the costs of rebellion”; their incomplete-information analysis (their Proposition 2) delivers a unique equilibrium in which the rebellion succeeds if and only if the fundamental is below a threshold, equal to $1 - c$ in their psychological rewards setting, where c is the cost of participation. They adopt “the standard conception of state repression as any action by the state that ‘raises the contenders’ cost of collective action’ (Tilly 1978, 100),” representing more repression as a higher c , and they add a first stage in which the government chooses its repression level by minimizing the sum of the regime change threshold and a convex cost of repressive capacity. A key ingredient of their psychological rewards setting, which the present model adopts, is that the participation reward is contingent on the movement’s success: building on Wood’s (2003) notion of pleasure-in-agency, participants enjoy the psychological reward of having taken part only if the rebellion succeeds, and this success-contingency is what generates the strategic complementarity that the global game resolves.

Little (2016) distinguishes two coordination problems that potential protesters face: a “political coordination” problem, deciding whether to participate without knowing how many others are dissatisfied enough to join, and a “tactical coordination” problem, deciding when, where, and how to protest. His political coordination model also features a continuum of citizens with heterogeneous assessments and equilibria in cutoff strategies, and he shows that better tactical information “almost always lowers the effective cost of protest, leading to larger (and more cohesive) antiregime action.” This distinction matters for the present model because the critical threshold citizens face here is precisely the kind of object about which communication technology conveys information, a point I return to in the empirical implications.

3 Modeling protests, repression, and policy change.

In this section, I develop a model of citizens who decide to protest to change a disliked policy. The citizens face a coordination and collective action problem: the protest successfully changes the policy only if citizens' participation reaches a given threshold. The threshold, moreover, is not common knowledge: it is an uncertain fundamental about which citizens hold noisy private assessments. On the other side, the state would prefer not to change the policy. To influence protest participation, it can set a repression severity policy that inflicts costs on individuals who decide to protest. Nevertheless, producing a given level of repression (i.e., a given number of victims of repression) is costly in resource utilization, and setting a severity of repression policy is costly also because of public opinion backlash consequences. In what follows, I begin by setting up the citizens' participation problem and its complete-information benchmark. Next, I solve the global game. Then, I propose the two-stage sequential game in which the state decides the severity of the repression policy first, and the strategic implications of the model are inspected.

3.1 The citizens' participation problem.

There is a continuum of citizens of mass one, indexed by $i \in [0, 1]$. Each citizen chooses between participating in the protest (P) or not (NP); let $n_i \in \{0, 1\}$ indicate participation and let $N^* \in [0, 1]$ denote the realized mass of participants. The objective of protesters is to change a policy that has such a degree of gridlock that it is tough to change over traditional forms of representation. The protest is successful if participation reaches a critical threshold τ : the policy changes if and only if $N^* \geq \tau$.

The payoff primitives are as follows. A change in the policy provides to all individuals in the society a collective and indivisible benefit of B_p (i.e., the policy change provides public good benefits). Individuals who participate may get an expressive or emotional benefit d_p ; for example, they may feel that fighting for a just cause is good in and of itself. Participants also bear the expected cost of repression $\gamma_r \cdot c_r$, where c_r is the severity of repression (the disutility inflicted on a repressed protester) and $\gamma_r \in [0, 1]$ is the probability that a participant is a victim of state repression. Both c_r and γ_r are policy variables for the state.

Two features of the continuum setting must be made explicit, because they discipline how the

primitives enter the participation margin.

First, no individual citizen is pivotal. In a finite-player version of this game, the term Δp_i , the change in the probability of reaching the threshold induced by one's own participation, would carry the public good benefit B_p into the individual calculus. In the continuum, a single citizen has measure zero, so $\Delta p_i = 0$: the citizen receives B_p when the protest succeeds whether or not she participated, and B_p cancels from the comparison between participating and abstaining. The free-riding logic is total here. This is not a defect of the machinery but a substantive statement it forces us to make: in a mass protest, the policy prize itself cannot be what moves the marginal participant, because her participation does not detectably move the prize.

Second, for participation to respond to the prospects of success at all, the expressive benefit must be success-contingent. I follow the psychological rewards setting of Bueno de Mesquita and Shadmehr (2023), which builds on Wood's (2003) pleasure-in-agency: the expressive reward d_p is enjoyed by a participant only if the movement succeeds. The satisfaction of having been part of forcing the change is realized when the change is forced; a participant in a failed protest bears the costs and receives no expressive payoff. This success-contingency is a substantive assumption, and it is what generates the coordination problem: were d_p unconditional, participation would reduce to the comparison of d_p against $\gamma_r c_r$, independent of any belief about others, and the strategic structure would disappear. An unconditional expressive reward is a defensible alternative reading; the success-contingent reading is the one under which mass participation responds to the prospects of success, which is the phenomenon this paper models.

With these two features, the net payoff of participating relative to abstaining, for citizen i , is

$$u_i(P) - u_i(NP) = \mathbb{1}\{N^* \geq \tau\} \cdot d_p - \gamma_r \cdot c_r,$$

since $B_p \cdot \mathbb{1}\{N^* \geq \tau\}$ enters both sides and cancels. A citizen therefore participates when her belief that the protest will succeed, times the expressive reward, exceeds the expected repression cost.

3.2 The complete-information benchmark.

Suppose first, as a benchmark, that τ is common knowledge. Assume $0 < \tau \leq 1$, the interesting range, and $\gamma_r c_r < d_p$, so that participation is worthwhile when success is assured. Then two pure

strategy equilibria coexist. If every citizen participates, $N^* = 1 \geq \tau$, the protest succeeds, and each participant receives $d_p - \gamma_r c_r > 0$; no citizen gains by deviating, since abstaining forfeits $d_p - \gamma_r c_r$ while leaving success (and hence B_p) unchanged. If no citizen participates, $N^* = 0 < \tau$, the protest fails, and a lone deviator would pay $\gamma_r c_r$ for a failed protest; so universal abstention is also an equilibrium. If instead $\gamma_r c_r > d_p$, universal abstention is the unique equilibrium: participation is dominated even under assured success.

This benchmark mirrors the complete-information benchmark of Bueno de Mesquita and Shadmehr (2023, their Proposition 1), and it inherits the two problems they identify: the multiplicity makes prediction difficult, and the equilibrium set is insensitive to the parameters, including the repression policy itself. One could respond by selecting an equilibrium, for instance the mixed strategy equilibrium, and reading its participation probabilities as frequencies. I instead take seriously that the critical threshold is a variable citizens cannot observe. Unlike in elections under plurality rule, where the threshold to win is common knowledge, the size a protest must reach to force a policy change is written nowhere, and how citizens form beliefs about τ is central to any behavioral prediction. That observation becomes the engine of the model.

3.3 The global game.

Let the critical threshold τ be drawn by nature and unobserved by citizens. Citizens share an improper uniform prior over $\mathbb{R}$, the standard device in this literature to isolate strategic uncertainty (Morris and Shin 2003; Bueno de Mesquita and Shadmehr 2023).¹ Each citizen receives a noisy private signal

$$x_i = \tau + \sigma \varepsilon_i,$$

where the ε_i are i.i.d. with a log-concave density f and c.d.f. F with full support on $\mathbb{R}$, independent of τ , and $\sigma > 0$ scales the noise. A citizen with a low signal believes the required crowd is small and success likely; a citizen with a high signal believes success is out of reach.

A strategy is a mapping from signals to participation. Since the participation payoff is decreasing in the (believed) threshold, it is natural to look for equilibria in cutoff strategies: participate if

¹The improper prior delivers exact closed forms. The same characterization obtains for any smooth proper prior in the limit as the signal noise becomes small.

and only if $x_i < \hat{x}$. Given such a cutoff, the mass of participants at true threshold τ is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ . The protest therefore succeeds if and only if τ is below a unique critical value $\hat{\tau}$, and the equilibrium is characterized by the same two conditions as in the canonical regime change game (Bueno de Mesquita and Shadmehr 2023, their equation (1)):

$$\text{(belief consistency / critical mass):} \quad F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = \hat{\tau},$$

$$\text{(indifference of the marginal citizen):} \quad \Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) \cdot d_p = \gamma_r c_r.$$

The first condition says that at the critical threshold the crowd is exactly critical. The second says that the citizen holding exactly the cutoff signal is indifferent between participating and abstaining. Under the improper uniform prior, the posterior of τ given $x_i = \hat{x}$ is $\hat{x} - \sigma \varepsilon$, so

$$\Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) = 1 - F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = 1 - \hat{\tau},$$

where the last equality substitutes the belief consistency condition. The indifference condition then collapses to $(1 - \hat{\tau})d_p = \gamma_r c_r$, and the equilibrium success threshold is

$$\hat{\tau}(c_r) = 1 - \frac{\gamma_r c_r}{d_p}, \quad \text{for } \frac{\gamma_r c_r}{d_p} \in [0, 1]. \quad (1)$$

This is the exact analog of the threshold $\theta^p = 1 - c$ in Bueno de Mesquita and Shadmehr (2023, their equation (2) and Proposition 2), with the normalized participation cost $\gamma_r c_r / d_p$ in the role of c . The equilibrium is unique in the class of monotone strategies, and standard arguments extend uniqueness beyond that class.² Three remarks.

First, the multiplicity is gone. For every value of the repression policy there is one prediction: the protest succeeds if and only if the realized threshold satisfies $\tau \leq \hat{\tau}(c_r)$. The comparative

²See Morris and Shin (2003) and the technical discussion in Bueno de Mesquita and Shadmehr (2023, their footnote 8).

statics of the model are therefore unconditional statements, not statements contingent on which equilibrium is played or on a selected mixed strategy.

Second, the repression policy now moves the outcome smoothly. The deterrence derivative is

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{\gamma_r}{d_p}.$$

Each unit of severity shrinks the set of fundamentals at which the protest succeeds at a constant rate, inversely proportional to the expressive reward. Repression against a strongly motivated citizenry (high d_p) is blunt; against a weakly motivated one it is sharp. This is the global games rendering of a familiar intuition: intrinsic motivation erodes the effectiveness of severity.

Third, the deterrence margin is governed by d_p alone, not by $d_p + B_p$. In a small-numbers setting where participation is pivotal, full deterrence would require severity to climb over the policy prize itself, since the pivotal citizen weighs B_p directly. Here, full deterrence ($\hat{\tau} = 0$) is reached already at $c_r = d_p/\gamma_r$: severity never needs to climb over B_p , because the free-riding of the continuum has already removed B_p from the participation margin. This is a substantive, testable implication of the mass-mobilization margin, discussed further below.

For the state's problem it is convenient to work in the small-noise limit $\sigma \rightarrow 0$. In that limit the cutoff $\hat{x}$ converges to $\hat{\tau}$ and participation becomes sharp: $N^*(\tau) \rightarrow 1$ for $\tau < \hat{\tau}$ and $N^*(\tau) \rightarrow 0$ for $\tau > \hat{\tau}$. Protests that succeed are massive; protests that fail do not materialize. This stark prediction is an artifact of vanishing noise, and with $\sigma > 0$ participation falls continuously in τ ; the limit is used only to keep the state's objective in closed form.

3.4 The state's problem.

The state moves first and sets the repression severity policy, anticipating the unique equilibrium of the citizens' stage. I fix $\gamma_r = 1$ to focus on c_r as the policy variable, and I assume the following cost function for the production of repression:

$$f(c_r, N^*) = \theta_0 \cdot N^* + \theta_1 c_r^2,$$

where θ_0 is the parameter that influences the marginal cost of repressing one additional protester (e.g., costs in repressive inputs or equipment) and θ_1 is the marginal cost of increasing the severity of repression (e.g., public opinion marginal costs). The state keeps B_s if the protest fails and loses it if the protest succeeds.

The state evaluates its policy *ex ante*, before nature draws τ . For this evaluation the improper prior is not usable, and I endow the state with a proper planning prior $\tau \sim U[0, 1]$, following the same logic by which Bueno de Mesquita and Shadmehr (2023) give their government a proper distribution over the fundamental at its investment stage.³ Under this prior and the small-noise limit, the probability that the protest succeeds is $\Pr(\tau \leq \hat{\tau}) = \hat{\tau}(c_r)$, and the expected mass of protesters is $E[N^*] = \hat{\tau}(c_r) \cdot 1 = \hat{\tau}(c_r)$, since in the limit the protest is total exactly when it succeeds. The state's expected utility is

$$u_S = (1 - \hat{\tau}(c_r))B_s - \theta_0 \hat{\tau}(c_r) - \theta_1 c_r^2 = \frac{c_r}{d_p} B_s - \theta_0 \left(1 - \frac{c_r}{d_p}\right) - \theta_1 c_r^2,$$

substituting equation (1) with $\gamma_r = 1$. The objective is strictly concave in c_r , so the first order condition characterizes the optimum on the relevant domain $c_r \in [0, d_p]$; beyond d_p , additional severity produces no further deterrence, since $\hat{\tau}$ is already zero, and only adds public opinion costs. The first order condition is

$$\frac{B_s + \theta_0}{d_p} = 2\theta_1 c_r,$$

which reads: the marginal benefit of severity, deterrence at rate $1/d_p$ valued at the status quo prize plus the production cost saved per deterred protest, equals the marginal public opinion cost. Solving,

$$c_r^* = \min \left\{ \frac{B_s + \theta_0}{2\theta_1 d_p}, d_p \right\}. \quad (2)$$

The interior solution obtains when $B_s + \theta_0 < 2\theta_1 d_p^2$; otherwise the state jumps to the full-deterrence corner $c_r^* = d_p$, at which the protest never succeeds. At the interior optimum, the equilibrium

³Their government minimizes $\theta^j(c) + C(c)$; the uniform planning prior makes the probability of regime change equal to the threshold itself, which is the same simplification used here.

success threshold, which is also the probability of policy change and the expected protest size, is

$$\hat{\tau}^* = 1 - \frac{B_s + \theta_0}{2\theta_1 d_p^2}.$$

3.5 Comparative statics.

The comparative statics at the interior solution are, without exception, unambiguous:

$$\begin{aligned} \frac{\partial c_r^*}{\partial B_s} &= \frac{1}{2\theta_1 d_p} > 0, & \frac{\partial c_r^*}{\partial \theta_0} &= \frac{1}{2\theta_1 d_p} > 0, & \frac{\partial c_r^*}{\partial \theta_1} &= -\frac{B_s + \theta_0}{2\theta_1^2 d_p} < 0, \\ \frac{\partial c_r^*}{\partial d_p} &= -\frac{B_s + \theta_0}{2\theta_1 d_p^2} < 0, & \frac{\partial c_r^*}{\partial B_p} &= 0. \end{aligned}$$

Figure 1 illustrates these comparative statics by region of the parameter space, plotting the latent interior solution against the effective policy in each regime.

The signs admit direct interpretation. An increase in the production cost parameter θ_0 raises the chosen severity: if repressing each protester is dearer, the state deters harder to face fewer of them. An increase in the public opinion cost parameter θ_1 lowers severity. An increase in the intrinsic motivation d_p lowers severity, because motivated citizens are expensive to deter; note, however, that here the state's retreat does not fully compensate, since $\hat{\tau}^*$ is increasing in d_p , so more motivated citizenries both face milder repression and succeed more often.

Two features of these results deserve emphasis.

First, the effect of the status quo prize B_s is unambiguously positive, with no parametric condition attached. In coordination models resolved by equilibrium selection, comparative statics of this kind are typically regime-conditional, holding in some regions of the parameter space and reversing in others. Here the unique threshold equilibrium makes the deterrence technology the linear relation (1), and the marginal value of severity is monotone in the stakes.

Second, B_p never enters the state's problem. The severity policy does not reckon with the policy prize at all: positive severity is optimal whenever $B_s + \theta_0 > 0$, that is, always, and the size of the prize the citizens would win is irrelevant to how hard the state represses. This is the model working as intended, and it delimits its scope. The model instantiates a mass setting in which individual participation is not pivotal; in a small-numbers setting, the pivotal calculus of a small

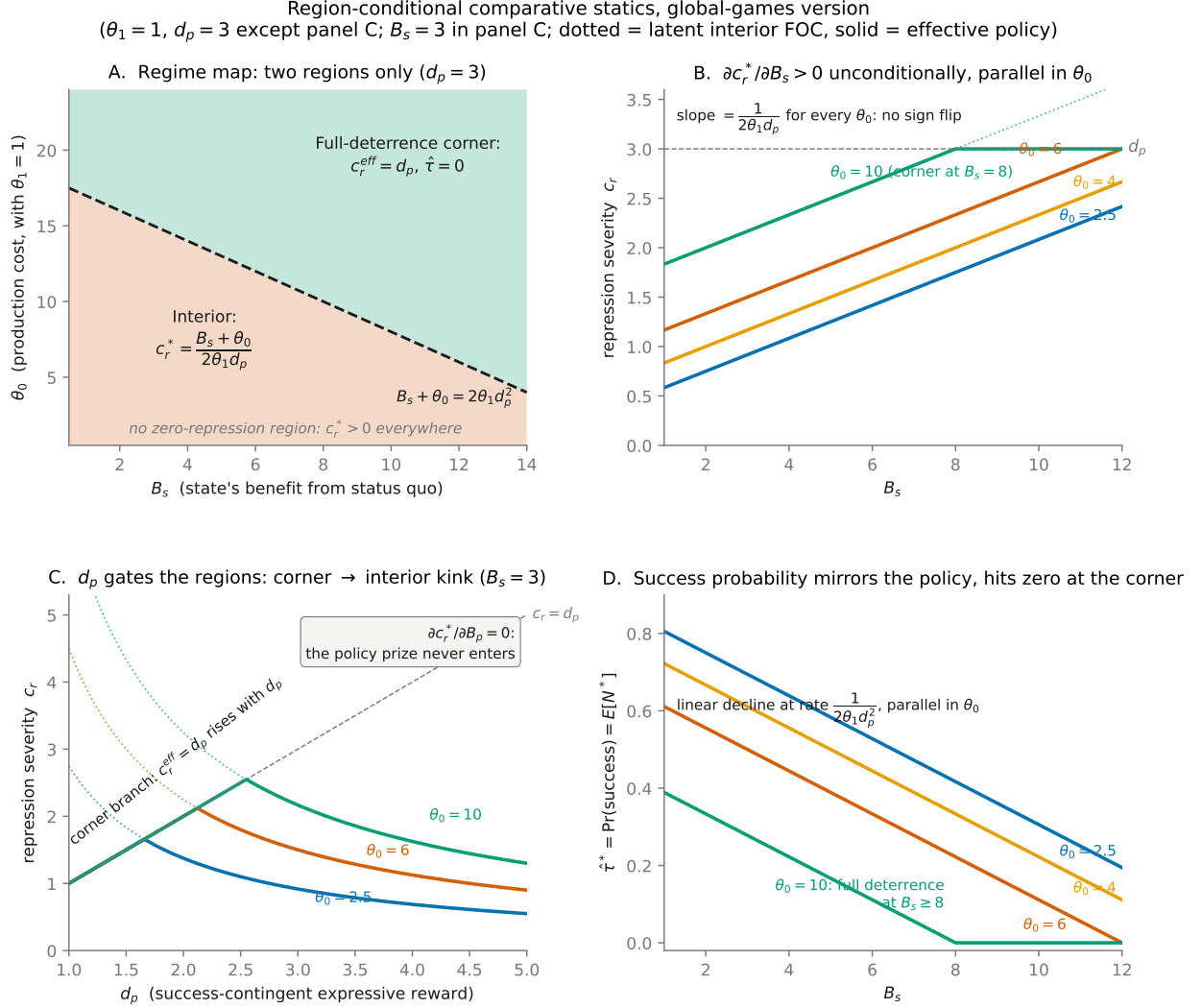


Figure 1: Region-conditional comparative statics of the repression severity policy ($\theta_1 = 1$; $d_p = 3$ except in panel C; $B_s = 3$ in panel C). Panel A maps the two regions of the parameter space: the interior solution and the full-deterrence corner, separated by $B_s + \theta_0 = 2\theta_1 d_p^2$; there is no zero-repression region. Panel B shows that c_r^* rises linearly in B_s with slope $1/(2\theta_1 d_p)$ for every θ_0 , flattening at the corner $c_r = d_p$. Panel C shows the kinked, non-monotone effect of d_p : the effective policy rides the corner $c_r = d_p$ upward while full deterrence is cheap, then switches to the declining interior solution. Panel D plots the equilibrium success probability $\hat{\tau}^*$, which declines linearly in B_s at rate $1/(2\theta_1 d_p^2)$ and reaches zero at the corner. Dotted curves are the latent interior first order condition; solid curves are the effective policy.

group carries B_p into the participation margin and, with it, into the severity policy. Which margin governs actual protest participation is an empirical question, and the sharp difference in the role of B_p across the two margins converts that question into a testable one.

3.6 An extension: Repression, protests, and emotions.

It is possible to adapt the model and introduce the role of emotions in the game's dynamics. Let the expressive reward be a function of the emotional state “ e ” of an individual, $d_p(e) = d_0 + e(c_r)$. To the extent that, in democracies, open media allows citizens to learn about how repressive is the average action of a state operative, citizens may get angered when they learn about a high degree of severity of repression, so that $e'(c_r) > 0$ for low levels of c_r ; yet for high or extreme levels of c_r , anticipatory fear may dominate and $e'(c_r) < 0$.

In the global games framework this channel acquires a precise transmission mechanism: the equilibrium threshold becomes

$$\hat{t}(c_r) = 1 - \frac{\gamma_r c_r}{d_0 + e(c_r)},$$

so the emotional response enters the citizens' stage through the denominator of the normalized participation cost. In what follows, I develop the extension in three steps. First, I derive the general condition under which severity deters or backfires. Next, I solve the state's problem when anger responds linearly to severity. Then, I consider an outrage specification in which the anger response accelerates, which produces a backfire region and caps the state's optimal severity.

Let us fix $\gamma_r = 1$ and differentiate the threshold with respect to the severity policy. We get

$$\frac{\partial \hat{t}}{\partial c_r} = - \frac{d_0 + e(c_r) - c_r e'(c_r)}{(d_0 + e(c_r))^2}.$$

Define the elasticity of motivation with respect to severity as $\varepsilon(c_r) = c_r e'(c_r)/d_p(c_r)$. Hence, severity deters ($\partial \hat{t}/\partial c_r < 0$) if and only if $\varepsilon(c_r) < 1$, and it backfires ($\partial \hat{t}/\partial c_r > 0$) if and only if $\varepsilon(c_r) > 1$. In other words, whether repression discourages or recruits is not a psychological primitive but an elasticity condition on the anger response. Note that this gives the “punishment puzzle” (Davenport 2007b) a precise formulation: the deterrence and backlash findings that coexist in the empirical literature (della Porta 1995; Christopher Michael Sullivan and Davenport 2017; Adam-

Troian, Çelebi, and Mahfud 2020) correspond to the two sides of the same inequality.

Let us begin with the linear case, $e(c_r) = \alpha c_r$ with $\alpha \geq 0$, so that $d_p(c_r) = d_0 + \alpha c_r$. Note that $\varepsilon(c_r) = \alpha c_r / (d_0 + \alpha c_r) < 1$ for every severity level, so linear anger never reverses deterrence; it dampens it, given that $\partial \hat{\tau} / \partial c_r = -d_0 / (d_0 + \alpha c_r)^2$. More importantly, the threshold is bounded below: as c_r grows, $\hat{\tau}$ approaches $1 - 1/\alpha$. Thus, if $\alpha \geq 1$, full deterrence is unattainable at any finite severity, and an angry-enough citizenry retains a strictly positive probability of policy change no matter how harsh the policy. If $\alpha < 1$, full deterrence requires $c_r = d_0 / (1 - \alpha)$, which grows without bound as α approaches one.

The state's problem remains strictly concave, and the first order condition becomes $(B_s + \theta_0)d_0 = 2\theta_1 c_r (d_0 + \alpha c_r)^2$, which has a unique positive solution.⁴ The full-deterrence corner is optimal if and only if $(B_s + \theta_0)(1 - \alpha)^3 \geq 2\theta_1 d_0^2$, which collapses to the condition of the baseline model at $\alpha = 0$ and becomes impossible to satisfy as α approaches one. Two comparative statics follow. First, $\partial c_r^* / \partial \alpha < 0$: facing a more anger-reactive citizenry, the state tempers the severity of repression. Second, $d\hat{\tau}^* / d\alpha > 0$: the equilibrium probability of policy change increases in anger-reactivity through both channels, directly, because anger blunts deterrence, and strategically, because the state represses less.

Next, let us consider an outrage specification, $e(c_r) = \alpha c_r^k$ with $k > 1$, in which the anger response accelerates with severity. The idea is that extreme severity is qualitatively different from routine crowd control: a police killing that saturates media coverage recruits participants in a way that tear gas does not (Reny and Newman 2021). Now $\varepsilon(c_r)$ is increasing in c_r and crosses one at $\tilde{c} = (d_0 / (\alpha(k - 1)))^{1/k}$, so the threshold $\hat{\tau}$ is U-shaped: severity deters below $\tilde{c}$ and backfires above it. Two consequences follow. First, the backfire point caps the state's policy, and the optimal severity never exceeds $\tilde{c}$, given that pushing beyond it raises both the probability of policy change and the costs of repression. Second, if $\hat{\tau}(\tilde{c}) = 1 - \tilde{c}(k - 1) / (d_0 k) > 0$, full deterrence is unattainable at any severity, and the best the state can do is push the probability of policy change down to its floor at the backfire point.

Finally, fear is the mirror case and requires little development. If $e'(c_r) < 0$, then $\varepsilon(c_r) < 0$, deterrence is amplified, and the qualitative structure of the baseline model is preserved with a

⁴The left-hand side is a positive constant and the right-hand side is strictly increasing from zero, so the two cross exactly once. The derivations for this section are collected in the appendix.

steeper deterrence derivative. The interesting asymmetry is that anger changes the structure of the state's problem, bounding its policy and possibly its payoff, while fear only rescales it. A full nonmonotone specification, in which anger dominates at low severity and fear at extreme severity, combines the two regimes; its analysis adds no closed forms and is left for future work.

4 Empirical implications.

The game-theoretic model developed in the latter section offers several empirical implications. In this section, I suggest some of them and propose ways in which we would try to estimate the empirical relation between protests and repression.

First, the model suggests that in any empirical estimation of either the effect of protests on repression or vice versa, a naive regression between the two will give wrong estimates, even if it has many meaningful controls to account for omitted variables. This happens because repression and the size of protests are simultaneously defined. Hence, empirical research should consider using structural estimation models or models that exploit some natural experiment, such as instrumental variables. The incomplete-information structure strengthens the case for the structural route: because the equilibrium is unique, the model delivers point predictions, a likelihood can be written down for observed outcomes, and the parameters $(d_p, \theta_0, \theta_1, B_s)$ are identified from variation in repression severity and protest success without an auxiliary equilibrium selection assumption. Multiplicity is precisely what makes structural estimation of coordination models treacherous, and it is what the machinery removes.

Second, one contribution of the model is to bring attention that, in democracies, the production of repression is costly. Even if we are unsure about the magnitude of public opinion costs, the level of overall repression has production costs such as spending on police trucks, ammunition, armored water cannon trucks, police office buildings, and others. To the extent that fiscal spending has opportunity costs, given that the incumbent politicians could have utilized such resources directly favoring voters, the degree of severity of repression will have nontrivial strategic considerations.

On the other hand, although it is not a particularly surprising result, an increase in public opinion marginal costs may reduce the severity of repression. Hence, if such an increase is in

place, as it is common in democracies, this could induce a bigger size of protests or more protests participation in general.

Moreover, suppose empirical researchers can track exogenous shocks that may influence either the production costs of the level of repression (in the model, θ_0) or shocks that influence the public opinion costs (in the model, θ_1), that are unrelated with the determinants of protest participation. Then, it should be possible to operationalize a measure of these shocks and exploit it under an instrumental variables model.

The model's distinctive predictions supply an additional, sharper test. The severity of repression should respond positively to the incumbent's stakes in the status quo (B_s) everywhere in the parameter space, and it should not respond to the size of the prize citizens would obtain from the policy change (B_p). The second prediction is the signature of the mass-mobilization margin: it holds because no participant is pivotal, and a pivotal calculus would restore B_p to the participation margin. Settings where protests are massive and individual pivotality is negligible should conform to this pattern; settings with small, organized, pivotal groups should not. A design comparing repression severity across policy disputes that differ in the dispersion of the prospective benefits, holding the incumbent's stakes fixed, would discriminate between the two margins.

The informational structure also generates implications of its own. Citizens' behavior depends on their beliefs about the critical threshold, and those beliefs respond to public information about expected turnout, past protest sizes, and logistical focal points. This connects directly to Little's (2016) distinction between political and tactical coordination: information that improves citizens' assessments of the crowd required, and of the crowd forthcoming, operates on the very fundamental τ of the present model, and his result that better tactical information systematically increases protest suggests that communication technology should compress the noise σ and sharpen the participation cascade around the threshold. Empirically, exogenous variation in connectivity or in media coverage of expected turnout could be used to test whether protest outcomes become more bimodal, either massive or absent, as the model predicts they should when strategic uncertainty shrinks.

However, one obvious and nontrivial challenge is how to measure any of these variables. First, although not easy to measure, protest size seems like the less difficult one of them all. Several data sets code protest event characteristics by analyzing media content. In some cases, protest

size participation is recorded.⁵ Additionally, using survey data, one can approximate the intrinsic motivations to participate in protests (d_p) at the individual level by using question items that gauge political involvement or attitudes towards political participation. Note that under the success-contingent interpretation adopted here, the relevant survey construct is not generic expressiveness but the anticipated satisfaction of having contributed to a victory, which is closer to efficacy and pleasure-in-agency item batteries. The variable B_p , a fundamentally unobserved construct and among the hardest in the model to measure, exits the participation margin here, which is a modest empirical convenience of this modeling strategy.

On the other hand, it may be possible to proxy for the variables that measure the state's marginal costs of repression. For example, one could measure the marginal costs of producing a level of repression (θ_0), measuring the average wage of the repressive operatives, or the percent of the GDP that the state devotes to repressive bureaus. Furthermore, we might use the distance of protest events to police stations (or the relevant repression bureau physical offices) as an instrument to the level of repression in a given protest event or episode. The idea is that if this distance is far, there is more cost in mobilizing the repressive operatives. Hence there is an incentive to increase the severity of repression. Following a repression severity policy defined at a high level, we would predict a smaller protest size.

On the other hand, the marginal costs related to public opinion may be proxied by the conditions of the competitive political environment. Yet, thinking about this variable should hint that we also must consider the institutional framework of any specific national or subnational political level. For example, if repressive operatives are managed at the subnational or local level, this could create different effects for incumbents at the national level. All in all, the idea is that if the relevant electoral setting is competitive, repression may be more costly for the incumbent.

Next, I briefly discuss ways in which we could measure the repression severity policy in democracies. In this case, data is indeed scarce. Although efforts are growing, such as the Mapping Police Violence project in the U.S., there is still much to be done. Yet, we still may want to think about how we could approach this construct if we had the data. At least, we could think of two ways to measure this. First, we could calculate the rate of victims of repression over the population

⁵For example, the Mass Mobilization Protest Data project, the Armed Conflict Location & Event Data Project, the Social Conflict Analysis Database, etc.

as some studies have done. The key open question in this approach would be how to choose the approach geographical area to define the base of the rate. Second, if we had such data, we could elaborate on a measure of physical or psychological trauma for the average victim of repression.

Finally, using survey data or experiments, it is possible to measure the relation between the severity of repression and the emotional state it elicits in citizens that may or may not plan to participate in future protests. One important aspect of such empirical assessment would be to test how different levels of repression severity may have nonlinear effects on the emotional state of individuals. For example, one may probe subjects with different and explicit consequences of repression victimization under a survey experiment. Then, one would measure their emotional state. Finally, one would ask individuals' dispositions or attitudes to participate in protests. Under the lens of the extension, the object to estimate is the elasticity of intrinsic motivation with respect to severity: the model predicts deterrence where it is below one and backlash where it is above one, so the design just described doubles as a direct measurement of the backfire point.

5 Conclusion.

Using a game-theoretic model, this paper explored the relationship between repression and protests motivated by policy change in democratic regimes. To accomplish this, a simple sequential game is elaborated that considers citizens' decisions to protest and the state's decision to set a policy of severity of repression to deter protest participation. The citizens' stage is developed as a global game of policy change: the critical threshold that protests must reach is an uncertain fundamental, citizens hold noisy private assessments of it, and the resulting incomplete information selects a unique equilibrium in threshold strategies where the complete-information benchmark exhibits multiple equilibria.

In this setting, the state moves first and defines a degree of severity of repression policy, which sets the individual costs for the repressed protester. The state would prefer that protests be minimal to keep the policy status quo unchanged, yet producing repression is costly for the state. In the second stage, citizens decide whether to protest under strategic uncertainty about the critical threshold. The protest succeeds, and the incumbent politicians are forced to change the policy, exactly when the realized threshold falls below an equilibrium cutoff that the state's severity policy

shifts down at a rate inversely proportional to citizens' intrinsic motivation.

The model offers several insights. First, the severity of repression and the prospects of protest are interlinked through a unique equilibrium, so the model's comparative statics are unconditional: severity rises with the incumbent's stakes in the status quo and with the marginal production cost of repression, and falls with the public opinion cost of severity and with citizens' intrinsic motivation. Second, the policy prize drops out of the participation margin altogether: in a mass protest no citizen is pivotal, so the prize cannot move the marginal participant. The contrast between this prediction and the pivotal logic of small-numbers settings is a testable difference between mass and small-numbers mobilization rather than a defect of the model. Third, increasing the marginal costs of the production level of repression increases the state's chosen degree of repression severity: deterrence economizes on the production of repression. On the other hand, increasing the marginal costs of negative public opinion due to repression makes the state reduce the severity of the repression policy.

A number of key open questions remain unexplored and suggest extensions to the model. First, the emotions extension develops the anger channel for linear and accelerating responses, showing that backlash is an elasticity condition and that outrage caps the state's optimal severity; a full nonmonotone specification, in which fear overtakes anger at extreme severity, remains open. Second, a repeated version in which citizens observe protest outcomes and update about a persistent threshold would connect the model to learning dynamics over protest waves. Third, the state itself could be given instruments over the information structure, censoring or publicizing turnout information, which the global games literature on communication technology suggests is a first-order policy margin (Little 2016). Overall, it seems that the biggest empirical challenge rests on how to measure key variables of the model. For example, one could measure the marginal production costs of the level of repression by measuring the average wage of repressive operatives or the percentage of GDP that the state allocates to repressive bureaus. On the other hand, the degree of severity of repression could be measured by the physical or psychological trauma of the average repressed protester. Nevertheless, it is likely that a more feasible option is just measuring the number of repressed protesters as a rate of the total local population. In any case, researchers should continue making efforts to understand repression and protests since these have become highly salient and influential phenomena in current politics, even in developed democracies.

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6 Appendix.

6.1 Derivation of the equilibrium threshold.

Fix a cutoff strategy profile in which every citizen participates if and only if $x_i < \hat{x}$. At true threshold τ , participation is $N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = F((\hat{x} - \tau)/\sigma)$, strictly decreasing and continuous in τ with range $(0, 1)$; hence there is a unique $\hat{\tau}$ with $N^*(\hat{\tau}) = \hat{\tau}$, and success ($N^*(\tau) \geq \tau$) occurs exactly on $\tau \leq \hat{\tau}$.

The marginal citizen holds signal $x_i = \hat{x}$. Under the improper uniform prior on τ , the posterior of τ given x_i is the distribution of $x_i - \sigma\varepsilon$ with $\varepsilon \sim f$; therefore

$$\Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) = \Pr\left(\varepsilon \geq \frac{\hat{x} - \hat{\tau}}{\sigma}\right) = 1 - F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = 1 - \hat{\tau},$$

using belief consistency $F((\hat{x} - \hat{\tau})/\sigma) = \hat{\tau}$ in the last step. Indifference of the marginal citizen requires her expected participation gain to be zero:

$$\Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) \cdot d_p - \gamma_r c_r = 0 \iff (1 - \hat{\tau})d_p = \gamma_r c_r \iff \hat{\tau} = 1 - \frac{\gamma_r c_r}{d_p}.$$

For $\gamma_r c_r \geq d_p$ participation is dominated for every signal and $\hat{\tau} = 0$; for $c_r = 0$, $\hat{\tau} = 1$. Uniqueness within monotone strategies follows from the strict monotonicity of the marginal citizen’s expected gain in $\hat{x}$; for the extension of uniqueness beyond monotone strategies under log-concave f , see Morris and Shin (2003) and the discussion in Bueno de Mesquita and Shadmehr (2023).

6.2 A caveat on the derivation of the equilibrium threshold.

A natural question about the derivation above is why the indifference condition is solved leaving $\hat{\tau}$ as is, instead of solving for the fixed point of the belief consistency condition. The answer is that

both conditions are used, and the fixed point is indeed solved; it simply delivers the unknown that the analysis does not display. Note that the equilibrium is a system of two equations in two unknowns, $(\hat{x}, \hat{\tau})$. The belief consistency condition alone cannot be solved as a standalone fixed point, given that it is one equation in two unknowns. In other words, for every candidate cutoff $\hat{x}$ it defines an induced threshold $\hat{\tau}(\hat{x})$, and the indifference condition provides the second relation; the equilibrium is the intersection of the two.

The reason the algebra collapses is that both conditions depend on $(\hat{x}, \hat{\tau})$ only through the statistic $z = (\hat{x} - \hat{\tau})/\sigma$. Under the improper uniform prior, the posterior probability that the protest succeeds is $\Pr(\tau \leq t \mid x_i = \hat{x}) = 1 - F((\hat{x} - t)/\sigma)$ for any evaluation point t . Hence, evaluating at $t = \hat{\tau}$ and substituting $F(z) = \hat{\tau}$ is not circular; it is the ordinary elimination of one unknown using the other equation. The system becomes triangular. First, the two conditions jointly yield $(1 - \hat{\tau})d_p = \gamma_r c_r$, which pins down $\hat{\tau} = 1 - \gamma_r c_r/d_p$. Then, the belief consistency condition, used again in the other direction, recovers the cutoff signal:

$$\hat{x} = \hat{\tau} + \sigma F^{-1}(\hat{\tau}).$$

This last expression is the fixed point in question. The main text does not display it because the state's problem only requires $\hat{\tau}$, and in the small-noise limit $\hat{x} \rightarrow \hat{\tau}$, so the distinction vanishes where the state's stage operates. Finally, note that this triangular structure is a consequence of the improper prior. Under a proper prior, the posterior does not reduce to $1 - F(z)$, the two conditions must be solved as a genuine two-dimensional fixed point, and no closed form obtains.⁶

6.3 Derivation of the state's optimum and comparative statics.

With $\gamma_r = 1$, the planning prior $\tau \sim U[0, 1]$, and the small-noise limit, $\Pr(\text{success}) = E[N^*] = \hat{\tau}(c_r) = 1 - c_r/d_p$ on $c_r \in [0, d_p]$. The state's objective is

$$u_S(c_r) = (1 - \hat{\tau})B_s - \theta_0 \hat{\tau} - \theta_1 c_r^2 = \frac{c_r}{d_p} B_s - \theta_0 \left(1 - \frac{c_r}{d_p}\right) - \theta_1 c_r^2.$$

⁶This is precisely why the improper prior is the standard device in this literature (Morris and Shin 2003; Bueno de Mesquita and Shadmehr 2023). At the critical threshold, the marginal citizen's belief about her rank in the population is uniform, so her success probability equals one minus the critical mass.

Then $u'_S(c_r) = (B_s + \theta_0)/d_p - 2\theta_1 c_r$ and $u''_S(c_r) = -2\theta_1 < 0$, so u_S is strictly concave and the maximizer on $[0, d_p]$ is the clipped first order condition

$$c_r^* = \min \left\{ \frac{B_s + \theta_0}{2\theta_1 d_p}, d_p \right\},$$

interior if and only if $B_s + \theta_0 < 2\theta_1 d_p^2$. Severity beyond d_p is never optimal: $\hat{\tau}$ is already zero there, so marginal deterrence is nil while the marginal public opinion cost $2\theta_1 c_r$ is strictly positive. Since $u'_S(0) = (B_s + \theta_0)/d_p > 0$, the optimum is strictly positive for all admissible parameters, with no further parametric condition.

At the interior optimum, direct differentiation of $c_r^* = (B_s + \theta_0)/(2\theta_1 d_p)$ gives

$$\frac{\partial c_r^*}{\partial B_s} = \frac{\partial c_r^*}{\partial \theta_0} = \frac{1}{2\theta_1 d_p} > 0, \quad \frac{\partial c_r^*}{\partial \theta_1} = -\frac{B_s + \theta_0}{2\theta_1^2 d_p} < 0, \quad \frac{\partial c_r^*}{\partial d_p} = -\frac{B_s + \theta_0}{2\theta_1 d_p^2} < 0,$$

and $\partial c_r^*/\partial B_p = 0$ since B_p does not enter the objective. The equilibrium success probability is $\hat{\tau}^* = 1 - c_r^*/d_p = 1 - (B_s + \theta_0)/(2\theta_1 d_p^2)$, whence

$$\frac{\partial \hat{\tau}^*}{\partial B_s} = \frac{\partial \hat{\tau}^*}{\partial \theta_0} = -\frac{1}{2\theta_1 d_p^2} < 0, \quad \frac{\partial \hat{\tau}^*}{\partial \theta_1} = \frac{B_s + \theta_0}{2\theta_1^2 d_p^2} > 0, \quad \frac{\partial \hat{\tau}^*}{\partial d_p} = \frac{B_s + \theta_0}{\theta_1 d_p^3} > 0.$$

The last derivative formalizes the double role of intrinsic motivation: a higher d_p lowers the chosen severity and raises the equilibrium probability of policy change.

6.4 Derivations for the emotions extension.

The elasticity condition. With $d_p(c_r) = d_0 + e(c_r)$ and $\gamma_r = 1$, differentiating $\hat{\tau}(c_r) = 1 - c_r/d_p(c_r)$ gives $\partial \hat{\tau}/\partial c_r = -(d_p(c_r) - c_r e'(c_r))/d_p(c_r)^2$. The numerator is positive if and only if $c_r e'(c_r)/d_p(c_r) < 1$, which is the elasticity condition $\varepsilon(c_r) < 1$ stated in the text.

Linear anger. With $e(c_r) = \alpha c_r$, the state's objective is $u_S(c_r) = (B_s + \theta_0)c_r/(d_0 + \alpha c_r) - \theta_0 - \theta_1 c_r^2$. Differentiating, $u'_S(c_r) = (B_s + \theta_0)d_0/(d_0 + \alpha c_r)^2 - 2\theta_1 c_r$ and $u''_S(c_r) = -2\alpha(B_s + \theta_0)d_0/(d_0 + \alpha c_r)^3 - 2\theta_1 < 0$, so the objective is strictly concave and the first order condition $(B_s + \theta_0)d_0 = 2\theta_1 c_r(d_0 + \alpha c_r)^2$ has a unique positive solution. For $\alpha < 1$, the full-deterrence corner sits at $c_r = d_0/(1 - \alpha)$, where $d_0 + \alpha c_r = d_0/(1 - \alpha)$; substituting into u'_S , the corner is optimal if and only if $(B_s + \theta_0)(1 - \alpha)^2/d_0 \geq 2\theta_1 d_0/(1 - \alpha)$, that is, $(B_s + \theta_0)(1 - \alpha)^3 \geq 2\theta_1 d_0^2$.

Comparative statics in α . Write the first order condition as $G(c_r, \alpha) = (B_s + \theta_0)d_0/(d_0 + \alpha c_r)^2 - 2\theta_1 c_r = 0$. Then $\partial G/\partial \alpha = -2(B_s + \theta_0)d_0 c_r/(d_0 + \alpha c_r)^3 < 0$ and $\partial G/\partial c_r = u''_S < 0$, so by the implicit function theorem $\partial c_r^*/\partial \alpha = -G_\alpha/G_{c_r} < 0$. For the equilibrium threshold, $d\hat{\tau}^*/d\alpha = \partial \hat{\tau}/\partial \alpha + (\partial \hat{\tau}/\partial c_r)(\partial c_r^*/\partial \alpha)$, where $\partial \hat{\tau}/\partial \alpha = c_r^2/(d_0 + \alpha c_r)^2 > 0$ and $\partial \hat{\tau}/\partial c_r = -d_0/(d_0 + \alpha c_r)^2 < 0$; both terms are positive, so $d\hat{\tau}^*/d\alpha > 0$.

Outrage. With $e(c_r) = \alpha c_r^k$ and $k > 1$, the numerator of $-\partial \hat{\tau}/\partial c_r$ is $d_0 + \alpha c_r^k - \alpha k c_r^{k-1} = d_0 - \alpha(k-1)c_r^{k-1}$, which is positive below and negative above $\tilde{c} = (d_0/(\alpha(k-1)))^{1/k}$; hence $\hat{\tau}$ is U-shaped with its minimum at $\tilde{c}$, where $\hat{\tau}(\tilde{c}) = 1 - \tilde{c}(k-1)/(d_0 k)$. For the cap on the optimal policy, write $u_S(c_r) = (B_s + \theta_0)(1 - \hat{\tau}(c_r)) - \theta_0 - \theta_1 c_r^2$; for $c_r > \tilde{c}$, the first term is strictly decreasing, given that $\hat{\tau}$ is increasing there, while the cost term is strictly increasing, so u_S is strictly decreasing beyond $\tilde{c}$ and the optimum satisfies $c_r^* \leq \tilde{c}$. All of the claims in this subsection were additionally verified numerically over randomized parameter draws.